
Problem Set 1

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PY 451 Section A1

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1 The Zero Vector

As requested, the numberings I am using are as given in the lecture notes. Let V be a vector space over $\mathbb{C}$ with zero vector $|0\rangle$. Then

$$0|v\rangle \stackrel{*}{=} (0+0)|v\rangle \stackrel{4}{=} 0|v\rangle + 0|v\rangle \stackrel{8,9}{\implies} 0|v\rangle + (0|v\rangle - 0|v\rangle) \stackrel{8}{=} 0|v\rangle \stackrel{*}{=} 0|v\rangle - 0|v\rangle \stackrel{8}{=} |0\rangle.$$

Note that the step denoted by $*$ follows by the field axioms, rather than the vector space axioms. Additionally, the step denoted $\star$ follows from the previous implication that $0|v\rangle + 0|v\rangle = 0|v\rangle$; I just moved stuff around to make it fit on one line.

2 Vector Spaces

For all of these problems, let V be the given set, and define addition and scalar multiplication pointwise; i.e. $(f+g)(x) = f(x) + g(x)$ and $(zf)(x) = z \cdot f(x)$ for all $f, g \in V$ and $z \in \mathbb{C}$.

1. Yes.

(a) Closure under addition:

$$(f+g)(0) = f(0) + g(0) = 0 + 0 = 0 \text{ and } (f+g)(L) = f(L) + g(L) = 0 + 0 = 0.$$

Therefore, $f+g \in V$;

(b) Commutativity (and associativity) of addition: Since $\mathbb{C}$ is commutative under addition, for all $x \in \mathbb{C}$,

$$(f+g)(x) = f(x) + g(x) = g(x) + f(x) = (g+f)(x).$$

Therefore, $f+g = g+f$. By the same logic, since $\mathbb{C}$ is associative under addition,

$$(f+(g+h))(x) = f(x) + (g(x) + h(x)) = (f(x) + g(x)) + h(x) = ((f+g)+h)(x).$$

Therefore, $(f+g)+h = f+(g+h)$, and V is associative under addition. $(f+g)(x) = f(x) + g(x) = g(x) + f(x) = (g+f)(x)$. Therefore, $f+g = g+f$. By the same logic since $\mathbb{C}$ is associative under addition, $(f+g)+h = f+(g+h)$;

(c) Note that the zero function $0(x) = 0$ is a function. For any $f \in V$,

$$(f+0)(x) = f(x) + 0(x) = f(x).$$

Therefore, $0(x)$ is an additive identity in V .

(d) Let $f \in V$. Define $-f$ as the function $(-f)(x) = -f(x)$. We have $f(x) + (-f)(x) = f(x) - f(x) = 0$ for all x . Therefore, $-f$ is an additive inverse of f .

(e) $(zf)(0) = z(f(0)) = z0 = 0$ and $(zf)(L) = z(f(L)) = z0 = 0$, so $zf \in V$;

2. No. Using the same definitions as before, note that the set has no additive identity. The zero function is the additive identity on the set of functions $[0, L] \rightarrow \mathbb{C}$:

$$(f + 0)(x) = f(x) + 0(x) = f(x) \implies \forall x : 0(x) = 0.$$

Since V is a subset of the set of maps $[0, L] \rightarrow \mathbb{C}$, we have that the 0 function must be the additive identity in V as well. We have $0'(x) = 0(x)$, and $0(0) = 0 \neq 0$. Therefore, $0 \notin V$.

3. Yes. It suffices to show the only element of the space is the 0 function. Let $f \in V$. Since f has a vanishing first derivative everywhere, it follows that f is continuous everywhere. Let $x \in [0, L]$ and suppose for contradiction that $f(x) > 0$. Since f is continuous on $[0, L]$, by the intermediate value theorem, f takes on every value on the interval $[0, f(x)]$ at some point on the interval $[0, x]$. Since f is continuous and $f(x) \geq 0$, we have that f is increasing on some subinterval of $[0, f(x)]$; equivalently, f has a positive first derivative on this interval. Therefore, $f(x) = 0$ for all $x \in [0, L]$. Note that this implies $f(L/2) = 0$, so that property is also satisfied. Therefore, $f \in V$ implies $f = 0$, so $V = \{0\}$ is the trivial vector space.

3 Eigenvalues

1. Notice that $M = |u\rangle\langle u|$. Let $\lambda|x\rangle = M|x\rangle$. We have

$$\lambda|x\rangle = |u\rangle\langle u|x\rangle = |u\rangle \sum_{n=1}^{10} nx_n.$$

We will use index notation to denote this as $|u\rangle nx_n$. We have

$$\lambda x_k = u_k nx_n = k nx_n.$$

We have

$$\lambda x_1 = nx_n.$$

Additionally,

$$\lambda x_k = k nx_n = k(\lambda x_1) \implies x_k = k x_1.$$

It follows that

$$\lambda x_1 = n(nx_1) = x_1 \sum_{n=1}^{10} n^2.$$

Therefore,

$$\lambda = \sum_{n=1}^{10} n^2 = 385.$$

Moreover, note that the rank of the matrix is 1:

```
1 #!/bin/python3
2
3 import numpy as np
4
5 u = np.array([i for i in range(1,11)])
6 print(np.linalg.matrix_rank(u @ u.T))
```

This program returns 1. The number of nonzero eigenvalues of a symmetric matrix is equal to its rank, and we implicitly assumed $\lambda \neq 0$ when we stated $x_k = kx_1$. If we suppose $\lambda = 0$, then we have $nx_n = 0$, since $k \neq 0$ for all k . We can define $x_k = (-1)^k 1/k$, and note that

$$\sum_{n=1}^{10} (-1)^n \frac{n}{n} = 5 - 5 = 0.$$

Therefore, 0 is also an eigenvalue for at least this singular vector.